

TCRBuilder: Multi-state T-cell receptor structure prediction

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Keywords:	Immunoinformatics, Protein structure prediction, Bioinformatics, Protein sequence analysis, Structural bioinformatics

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3 Dear Professor Elofsson,
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6 We would like to thank all three reviewers for their comments. We feel they made a positive
7 contribution to improving our manuscript and below is a point-by-point response.
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9 Format:
10 *Italicised paragraphs* are reviewers' comments and questions.
11 Normal fonts are author's response.
12 **Bolded fonts** are for the section headings under which the text are adjusted in response to
13 the reviewers' comments.
14 **Texts coloured in red** are the text changes made to the manuscript and Supplementary
15 Materials.
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Response to Reviewer: 1	
<i>Comments to the Author</i> <i>The manuscript introduces a web-server for building model structures of T-Cell Receptors (TCRs) from the sequences of their two protein chains. Comparison to existing web-servers shows similar accuracy (but see below). And I believe that a novelty here is the ability to suggest multiple conformations (but see below).</i>	
Author's response: We thank the reviewer for the clear description of our work.	
<i>1. There is so much polymorphism in TCRs – they are similar to germline antibodies in this. The fact that template free modeling was used for 2/3 of the targets shows that in real life scenario the majority of the models would not be accurate (template-free modeling is as good as guesswork for loops). So, who would use this tool and to what end? The authors should comment on this in the main text.</i>	
Author's response: Multiple existing TCR modelling tools (Klausen <i>et al.</i> , 2015; Gowthaman and Pierce, 2018; Schmitt <i>et al.</i> , 2019) have been used to study many aspects of TCR biology including the interactions between the peptide-major histocompatibility complex and TCRs (e.g. Jensen <i>et al.</i> , 2019; Lanzarotti <i>et al.</i> , 2019). This is the first attempt to develop a TCR modelling tool that is able to model flexibility, which is critical to TCR binding analysis. We agree with the reviewer that the method will not currently give accurate predictions every time, but as more structural data becomes available, template-based prediction coverage will increase. We have now made this clearer in the conclusion:	

Conclusion

TCRBuilder is the first TCR modelling server that captures the multiple stable conformations in TCR CDRs. The multi-state models it generates better represent the possible binding configurations a TCR can adopt, paving the way to understand polyspecificity in TCRs.

We have now also added the comments to the abstract.

Abstract

This enables the analysis of structurally-driven polyspecificity in TCRs, which is not possible with existing tools.

2. Figure 2B is very unclear. Maybe increase the size? Improve the graphical quality? Both?

Author's response:

We agree with the reviewer and have now updated the figure.

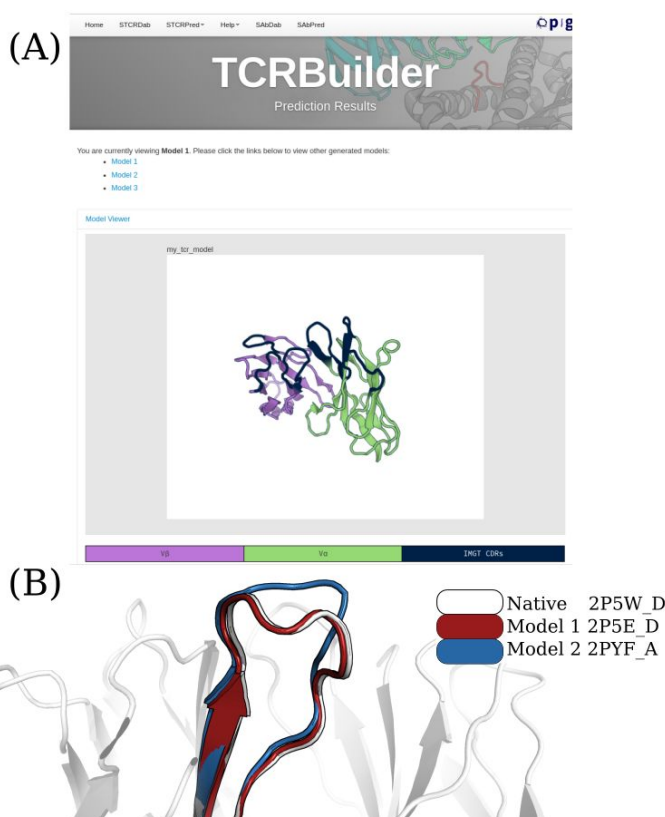


Fig. 1. (A) An example result page of the TCRBuilder web server. (B) An example case where a CDR α 3 sequence (AVRPLLDGTYIPT) with multiple distinct conformations was modelled. One of the conformations (2P5E chain D in red) was nearly identical to the native 2P5W chain D structure, but an alternative conformation of this loop (2PYF chain A in blue) is also predicted as a possible shape.

3. *“We compared the modelling performance of LYRA, TCRModel and Repertoire Builder”: References should be added to LYRA and the others.*

Author’s response:
These were referenced at the first mention in the paper as is standard practice:

1 Introduction

Several packages exist to model the structures of TCRs including LYRA, TCRModel and Repertoire Builder (Klausen *et al.*, 2015; Gowthaman and Pierce, 2018; Schritt *et al.*, 2019).

But as requested by the reviewer, we have now referenced them in this sentence as well.

2.1 Performance analysis

We compared the modelling performance of LYRA, TCRModel and Repertoire Builder with TCRBuilder (Klausen *et al.*, 2015; Gowthaman and Pierce, 2018; Schritt *et al.*, 2019), preventing them from using any structures released since 24 July 2019.

4. *“To test whether TCRBuilder is comparable to current state-of-the-art methods”: If these are web-servers, the reader should know that because they are readily available like TCRBuilder.*

Author’s response:
We have changed the wording in the main text to reflect the availability of these methods.

2.1 Performance analysis

To test whether TCRBuilder is comparable to current state-of-the-art, freely-available TCR web servers, we constructed a blind test set of the 10 (9 non-redundant) $\alpha\beta$ TCR structures released between 24 July 2019 and 22 October 2019.

5. *Since the performance of CTRBuilder is comparable to existing alternatives is there a reason to use it? Maybe that it provides multiple possible conformations?*

Author’s response:
As the title and conclusion stated, the novelty of our tool is to provide multiple conformations of the TCRs. We have now made this message clearer in the abstract.

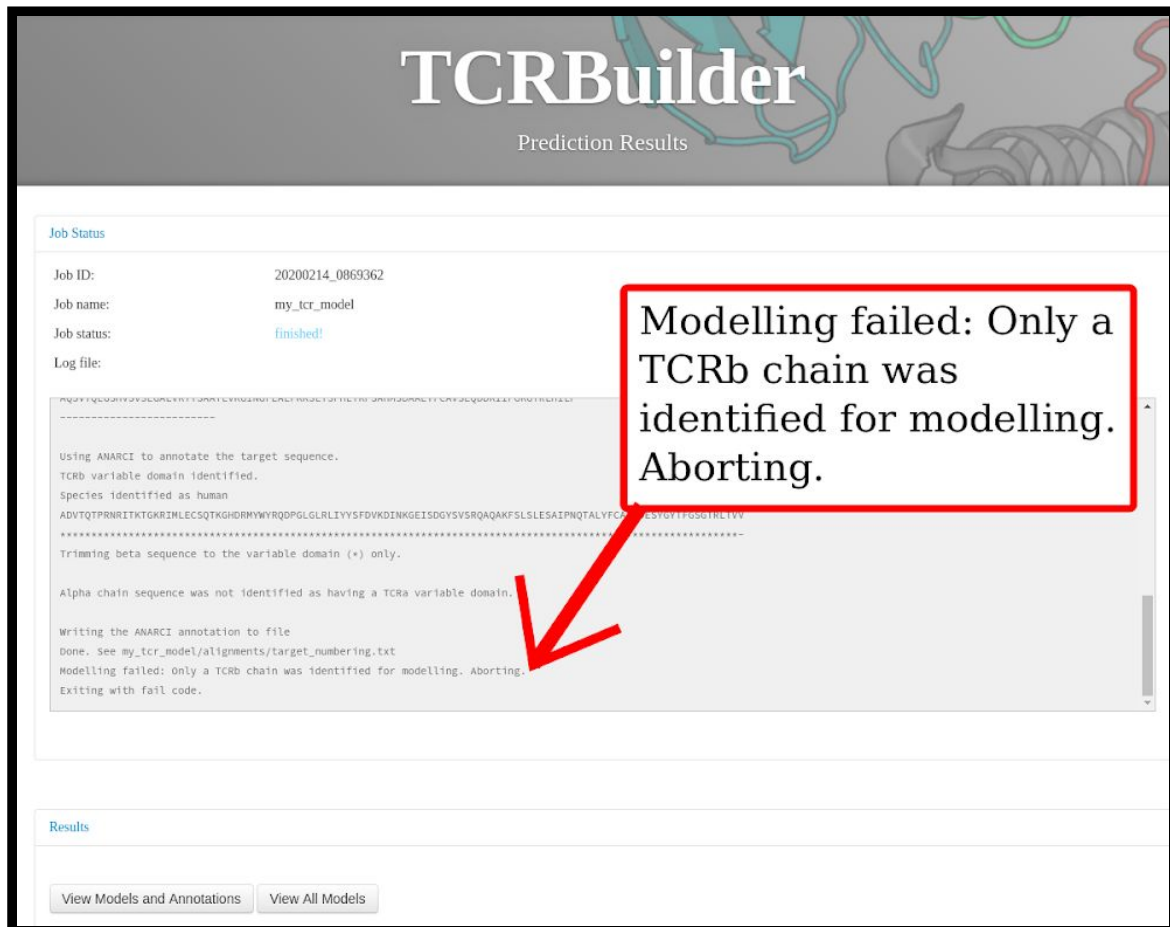
Abstract

This enables the analysis of structurally-driven polyspecificity in TCRs, which is not possible with existing tools.

6. I managed to run the example on the web-server. However, other cases didn't work for me. Presumably because I didn't upload legitimate TCRs. It would be helpful if the web-server included explanations about what kind of input is expected. Maybe a link to a CTRs database?

Author's response:

If a user fails to upload a pair of legitimate TCR sequences, the numbering step fails. This error message is recorded in the log file on the results page as shown:



In this instance, we entered an illegitimate α -chain sequence. The error message thus reads, "Modelling failed: Only a TCRb chain was identified for modelling. Aborting.".

If the reviewer encountered a different response from the webserver, we would like to know the input if possible.

We have also added further instructions on how to use the tool to the main page.

7. It would be nice if the output highlighted the CDRs. Both in the sequence and model structures.

Author’s response:
This functionality is already available as the default colouring scheme for visualising the sequences and structures:

HomeSTCERDabSTCERPredHelpSAbDabSAbPred

Model 1

View Model

Fv Region	Template PDB (chain)	Selection Method	Score
Alpha framework	6et8 (A)	sequence identity	0.97
CDR A1	6et8 (A)	TCR CDR-specific fread	46
CDR A2	6et8 (A)	TCR CDR-specific fread	59
CDR A3	3qeu (D)	TCR CDR-specific fread	25
Beta framework	4x6b (B)	sequence identity	0.65
CDR B1	5yxn (B)	TCR CDR-specific fread	27
CDR B2	3qdm (E)	TCR CDR-specific fread	31
CDR B3	5jzi (E)	sequence similarity	1
VA-VB orientation	6fr9 (BA)	Chosen by TRangle	1.00

Target/Template Sequence Alignment

Alpha chain target-templates alignment

IMGT	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
target	A	Q	S	V	T	Q	L	G	S	H	V	S	V	S	E	G	A	L	V	L	L	R	C	N	Y	S	S	S	V	P
framework	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
cdra1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
cdra2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
cdra3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	

Beta chain target-templates alignment

IMGT	-	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	37
target	-	A	D	V	T	Q	T	P	R	N	R	I	T	K	T	G	K	R	I	M	L	E	C	S	Q	T	K	G	H	D
framework	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
cdrb1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
cdrb2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
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MatchMismatchGapCDR

[Home](#)
[STCRDab](#)
[STCRPred](#)
[Help](#)
[SAbDab](#)
[SAbPred](#)

- Model 1
- Model 2
- Model 3
- [View All Models](#)

Model Viewer

my_tcr_model

Vβ

Vα

IMGT CDRs

Display options:

☐ Spacefill
 ☐ Wire
 ☐ Ball&stick
 ☒ Cartoon
 ☐ Spin on/off

Annotation options:

☒ IMGT CDRs
 ☐ Secondary structure
 ☐ Solvent exposed
 ☐ Domains

8. *“We evaluated all predicted models against all native structures with the same sequence, and select the best model for each structure”: How is “best” defined? Hopefully not the conformation which is closest to the native structure that you try to reproduce. Otherwise TCRBuilder may be inferior compare to the other web-servers.*

Author’s response:
We thank the reviewer for the comment and apologise for the confusion. For TCR sequences which display multiple conformations and for which we built multiple conformations when scoring, we assessed the performance with the “conformer” model that most closely replicates the native structure. We have now changed the text in the Supplementary Materials to clarify this concept.

Supplementary Results
Cross-validation performance

We compared the performance of the “conformer” model described below, to the native structure.

We assessed if any of the models in the ensemble capture a conformation that closely approximates the native structure. Hence, in the performance evaluation, we reported the lowest RMSD attained by the ensemble, to each of the native structures in the sequence-redundant set. The selected model is referred to as the “conformer” model.

9. *Same questions about other cases where the description refers to “best”. For example, the headers of Fig. S2.*

Author’s response:
This definition has now been added to the Supplementary Materials.

Supplementary Results
Cross-validation performance

We assessed if any of the models in the ensemble capture a conformation that closely approximates the native structure. Hence, in the performance evaluation, we reported the lowest RMSD attained by the ensemble, to each of the native structures in the sequence-redundant set. The selected model is referred to as the “conformer” model.

10. The legend to Figure S2 should be expanded. For example, what do the box plots for “CDRs” and “Total” measure?

Author's response:

We thank the reviewer for the comment and apologise for the lack of clarity. The description has now been updated.

Supplementary Results

Figure S2. The root-mean-square-distance (RMSD) of all backbone atoms across the respective regions, between the native structure and the model with the lowest RMSD. The RMSD for “CDRs” is calculated by dividing the backbone RMSD in all CDRs divided by the total lengths of all CDR sequences; whereas the RMSD for “Full” is calculated over the framework and CDR regions.

11. “We tested LYRA and Repertoire Builder”: References to these tools should be added.

Author's response:

We thank the reviewer for pointing this out and we have now added the references in the Supplementary Materials.

Supplementary Results

Blind set performance

We tested LYRA and Repertoire Builder using their normal modes (Klausen *et al.*, 2015; Schmitt *et al.*, 2019). TCRModel provides two modes for users. The results with no refinement are called “(norefine)”, and the ones with refinement are referred to as “(refine)” (Gowthaman and Pierce, 2018).

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Response to Reviewer: 2
2) b) <i>the references are somehow self-centered; relevant literature is missing, e.g., Crooks et al. Frontiers in Molecular Biosciences 2018</i> Author's response: This is a short Applications Note limited to two pages. We are therefore unable to cite all of the literature that might be pertinent (e.g. Wu <i>et al.</i>, 2002; Krogsaard <i>et al.</i>, 2003; Armstrong <i>et al.</i>, 2008; Scott <i>et al.</i>, 2011; Hawse <i>et al.</i>, 2014; Rossjohn <i>et al.</i>, 2015; Ayres <i>et al.</i>, 2016; Bello <i>et al.</i>, 2016; Holland <i>et al.</i>, 2018). In the case of Crooks <i>et al.</i>, 2018, this paper is in the area of TCR CDR conformational flexibility where there are many papers describing this property. In that area, we selected to cite the earliest work (Garcia <i>et al.</i>, 1998), a review (Morris <i>et al.</i>, 2012) and the most recent work which is our own paper. A more thorough appraisal has been summarised and reviewed in earlier papers. We would ask the editor if it would be appropriate to extend the list of reference within the space constraints of this manuscript.

Response to Reviewer: 3*Comments to the Author*

The Application Note "TCRBuilder: Multi-state T-cell receptor structure prediction" presents a web-based tool to predict the multi-state TCR structures given the potential of multi-conformations of the binding site. It's quite beneficial for such a tool to consider structural variability as it leads to polyspecificity.

Author's response:

We thank the reviewer for the clear description of our work.

A) Major:

1) It looks like most of this project is based on many earlier published projects, such as the databases were already published, e.g. Leem et al., 2017; Deane and Blundell, 2001; Choi and Deane, 2010; Lefranc et al., 2014. The modeling pipeline of TCRBuilder itself is largely based on the ABodyBuilder pipeline (Leem et al. 2016). May the authors specify the novel parts, which are sufficient enough for a new publication in this journal?

Author's response:

The algorithmic components here are, as the reviewer suggests, based on our earlier work. As this is an Applications Note, we believe that the novelty of using these components to generate an alternative TCR modelling pipeline, that provides the unique functionality of returning multiple conformations of loops is sufficient.

The benchmark results show that the TCRBuilder pipeline gives comparable performance, meaning that the way we connected the building blocks was sensible and useful. The ability to return multiple models is important for TCR modelling as it can point to how polyspecificity arises.

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2) *Where is the source code? The source code is critical to demonstrate innovation and workload.*

Author's response:
We decided to make TCRBuilder available as a webserver because of the direct link to our auto-updating STCRDab, a TCR structural database. The standard practice of *Bioinformatics* Applications Note submission is to release at least one of: webserver, source code or binary. Some examples with only a webserver available are listed below:

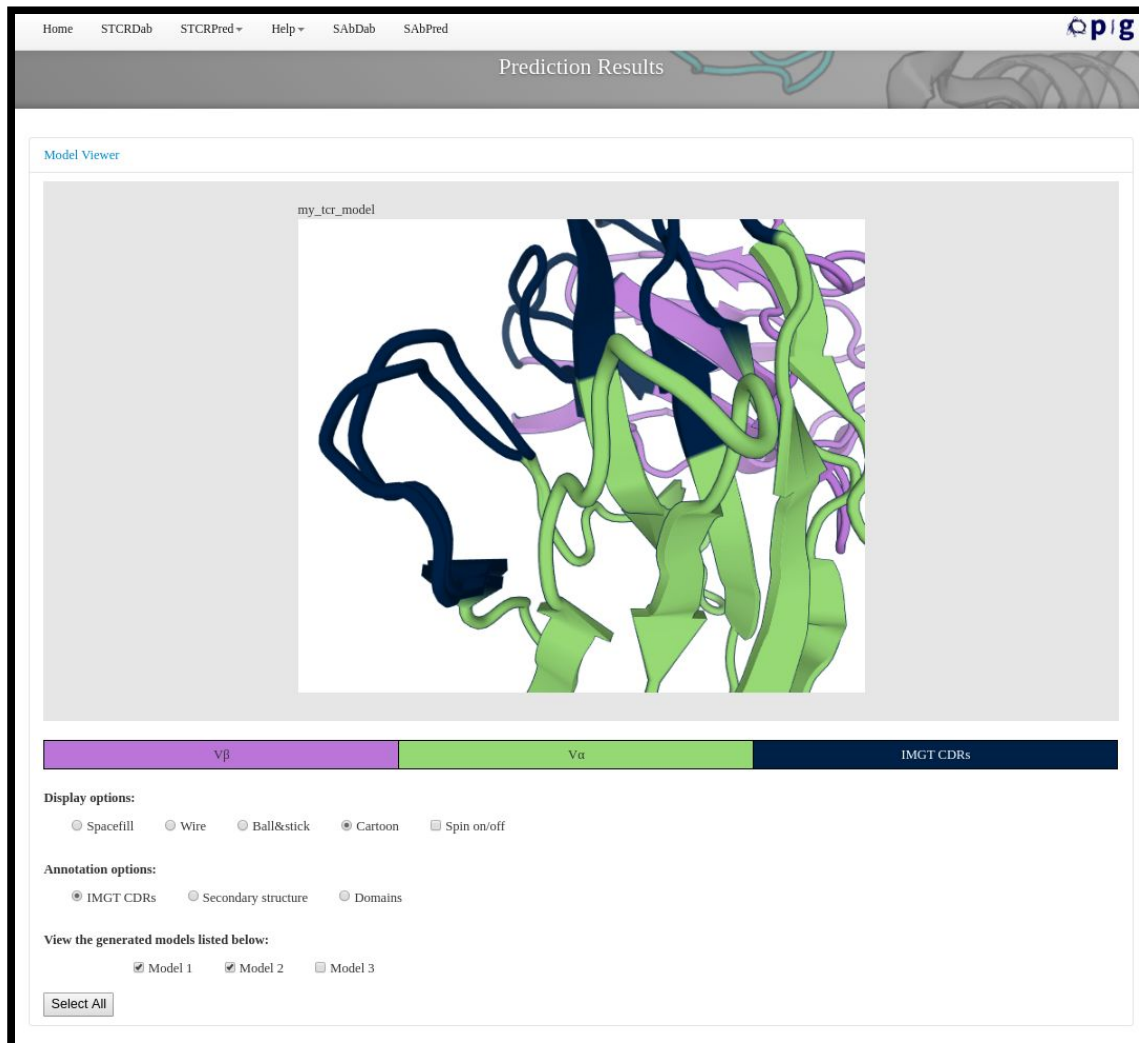
- Meng, G., Tariq, M., Jain, S., Elmetwaly, S. and Schlick, T. (2020). RAG-Web: RNA structure prediction/design using RNA-As-Graphs, *Bioinformatics*, **36**(2), 647–648.
- Ponti, R.D., Armaos, A., Vandelli, A. and Tartaglia, G.G. (2020). CROSSalive: a web server for predicting the *in vivo* structure of RNA molecules, *Bioinformatics*, **36**(3), 940-941.
- Lee, A. and Kim, D. (2020). CRDS: Consensus Reverse Docking System for target fishing, *Bioinformatics*, **36**(3), 959-960.

In addition, all other TCR modelling methods are also only available as webserver, with no source code freely-available. However we do hope to be able to release our code in the near future.

3) I tested the webapp in <http://opig.stats.ox.ac.uk/webapps/stcrpred/TCRBuilder/>. It's easy to use. However, it lacks an important function to overlap the different conformations predicted, although such overlap was shown in Figure 1 (B). May the authors add such functionality to highlight the different variable fragments which might be the most interesting parts?

Author's response:

Thank you for the suggestion. This has now been implemented on the webserver. An example output is given below:



We have rewritten the manuscript and the Supplementary Materials according to reviewers' comments, and would like to re-submit the manuscript for your consideration.

Kind regards,

Wing Ki Wong, Claire Marks, Jinwoo Leem, Alan P. Lewis, Jiye Shi and Charlotte M. Deane

Reference

- Armstrong, K.M., Piepenbrink, K.H. and Baker, B.M. (2008). Conformational changes and flexibility in T-cell receptor recognition of peptide–MHC complexes. *Biochemical Journal*, **415**(2), 183-196.
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- Scott, D.R., Borbulevych, O.Y., Piepenbrink, K.H., Corcelli, S.A. and Baker, B.M. (2011). Disparate degrees of hypervariable loop flexibility control T-cell receptor cross-reactivity, specificity, and binding mechanism. *Journal of Molecular Biology*, **414**(3), 385-400.

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3 Wu, L.C., Tuot, D.S., Lyons, D.S., Garcia, K.C. and Davis, M.M. (2002). Two-step binding
4 mechanism for T-cell receptor recognition of peptide–MHC. *Nature*, **418**(6897),
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Structural bioinformatics

TCRBuilder: Multi-state T-cell receptor structure prediction

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Abstract

Motivation: T-cell receptors (TCRs) are immune proteins that primarily target peptide antigens presented by the major histocompatibility complex. They tend to have lower specificity and affinity than their antibody counterparts, and their binding sites have been shown to adopt multiple conformations, which is potentially an important factor for their polyspecificity. None of the current TCR modelling tools predict this variability which limits our ability to accurately predict TCR binding.
Results: We present TCRBuilder, a multi-state TCR structure prediction tool. Given a paired $\alpha\beta$ TCR sequence, TCRBuilder returns a model or an ensemble of models covering the potential conformations of the binding site. This enables the analysis of structurally-driven polyspecificity in TCRs, which is not possible with existing tools.
Availability: <http://opig.stats.ox.ac.uk/resources>
Contact: deane@stats.ox.ac.uk
Supplementary information: Supplementary data are available at *Bioinformatics* online.

1 Introduction

In the immune system, antigens are foreign molecules recognised by two types of receptors with similar architectures: antibodies and T-cell receptors (TCRs). In humans, paired $\alpha\beta$ TCRs dominate the TCR population. They consist of one α and one β chain, analogous to the light-heavy chain dimer in antibodies. These four chains are encoded by the repertoire V, D and J genes of the respective TCR or antibody loci, and share a similar β -sandwich fold (Dunbar *et al.*, 2014). Despite the similar genetic formation and architecture, their biological functions differ: TCRs bind to peptide antigens presented by the major histocompatibility complex (MHC), and their binding is known to be polyspecific (Rudolph *et al.*, 2006). Antibodies on the other hand can bind to virtually any antigen with high specificity and affinity (Sela-Culang *et al.*, 2013). This distinction in binding preferences may be related to differences in their binding site structures (Wong *et al.*, 2019). For both TCRs and antibodies, the majority

of the binding site is made up of six loops known as the complementarity-determining regions (CDRs). It has been shown that when multiple solved structures of an identical sequence TCR CDR are available, 20% adopt distinct conformations, which is far higher than the ~8% observed in antibodies despite the far larger number of antibody structures (Wong *et al.*, 2019). This structural variability may allow alternative binding configurations for a given TCR, leading to its polyspecificity (Garcia *et al.*, 1998; Morris and Allen, 2012; Wong *et al.*, 2019). Several packages exist to model the structures of TCRs including LYRA, TCRModel and Repertoire Builder (Klausen *et al.*, 2015; Gowthaman and Pierce, 2018; Schmitt *et al.*, 2019). However, none of these tools produce predictions that reflect the multiple conformations of TCR CDRs. Here, we present TCRBuilder, a multi-state TCR modelling server. TCRBuilder is based on the ABodyBuilder pipeline for antibody modelling (Leem *et al.*, 2016) and predicts TCR models with multiple CDR conformations where applicable.

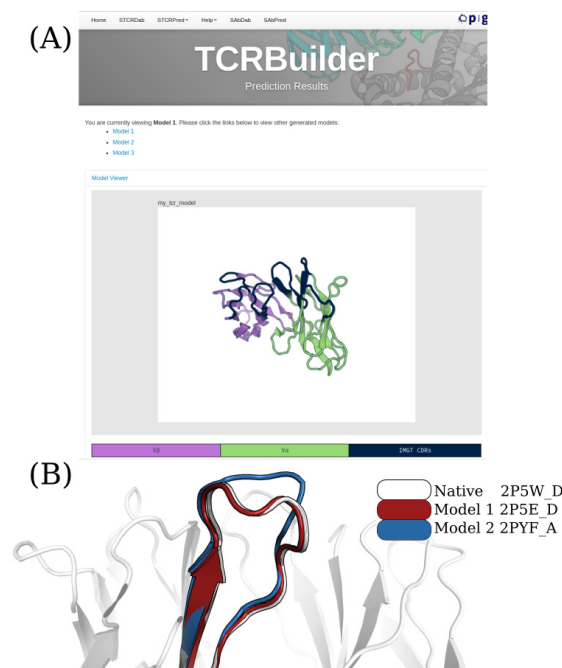


Fig. 1. (A) Input and an example result page of the TCRBuilder web server. (B) An example case where a CDR α 3 sequence (AVRPLLDGTYIPT) with multiple distinct conformations was modelled. One of the conformations (2P5E chain D in red) was nearly identical to the native 2P5W chain D structure, but an alternative conformation of this loop (2PYF chain A in blue) is also predicted as a possible shape.

2 Webserver

TCRBuilder is available as a web server (Fig. 1A). It takes as input a paired $\alpha\beta$ TCR sequence, and returns model structures with simple biophysical property annotations similar to those in ABodyBuilder (Leem et al., 2016). The detailed pipeline is described in the Supplementary Materials.

2.1 Performance analysis

To evaluate TCRBuilder, we carried out a leave-one-out cross-validation on 169 non-redundant TCR sequences in STCRDab (Leem et al., 2018), barring the algorithm from using full sequence-identical template structures. The performance was assessed by superimposing the predicted model onto the native structure, and calculating the root-mean-square-distance (RMSD) over the backbone atoms of the given region. RMSDs over the individual CDRs were obtained after aligning two anchor residues before and two after the CDR. If there were multiple crystal structures or models for a given sequence, all measurements were recorded. The average RMSD for the complete model was 1.49 Å. In the models, CDRs 1 and 2 on both the α and β chains are well predicted, but CDR3 less so. Of the 169 models generated, 106 required template-free modelling (Marks et al., 2017) for at least one of their CDRs. This is a far larger percentage than the <4% of antibody cases that required template-free modelling in a large test (Leem et al., 2016). These results highlight the lack of structural data available for TCRs.

To test whether TCRBuilder is comparable to current state-of-the-art methods, freely-available TCR web servers, we constructed a blind test set of the 10 (9 non-redundant) $\alpha\beta$ TCR structures released between 24 July 2019 and 22 October 2019. We compared the modelling performance of LYRA, TCRModel and Repertoire Builder with TCRBuilder (Klausen et al., 2015; Gowthaman and Pierce, 2018; Schmitt et al., 2019), preventing them from using any structures released since 24 July 2019.

TCRBuilder has comparable performance with existing TCR modelling tools, on this set of unseen structures (Supplementary Materials).

2.2 Predicting multiple conformations

As TCR CDR sequences are structurally variable, TCRBuilder is designed to capture the distinct conformations of CDRs that are known to exist (Wong et al., 2019). Eighty-nine of the 169 evaluated models are predicted to have more than one conformation of their binding sites. Figure 1B shows an example of a CDR α 3 sequence that can have multiple distinct conformations. As the number of TCR structures increases, it is likely that we will find many more CDR sequences with variable conformations.

3 Conclusion

TCRBuilder is the first TCR modelling server that captures the multiple stable conformations in TCR CDRs. The multi-state models it generates better represent the possible binding configurations a TCR can adopt, paving the way to understand polyspecificity in TCRs.

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Structural bioinformatics

TCRBuilder: Multi-state T-cell receptor structure prediction

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Abstract

Motivation: T-cell receptors (TCRs) are immune proteins that primarily target peptide antigens presented by the major histocompatibility complex. They tend to have lower specificity and affinity than their antibody counterparts, and their binding sites have been shown to adopt multiple conformations, which is potentially an important factor for their polyspecificity. None of the current TCR modelling tools predict this variability which limits our ability to accurately predict TCR binding.

Results: We present TCRBuilder, a multi-state TCR structure prediction tool. Given a paired $\alpha\beta$ TCR sequence, TCRBuilder returns a model or an ensemble of models covering the potential conformations of the binding site. This enables the analysis of structurally-driven polyspecificity in TCRs, which is not possible with existing tools.

Availability: <http://opig.stats.ox.ac.uk/resources>

Contact: deane@stats.ox.ac.uk

Supplementary information: Supplementary data are available at *Bioinformatics* online.

1 Introduction

In the immune system, antigens are foreign molecules recognised by two types of receptors with similar architectures: antibodies and T-cell receptors (TCRs). In humans, paired $\alpha\beta$ TCRs dominate the TCR population. They consist of one α and one β chain, analogous to the light-heavy chain dimer in antibodies. These four chains are encoded by the repertoire V, D and J genes of the respective TCR or antibody loci, and share a similar β -sandwich fold (Dunbar *et al.*, 2014). Despite the similar genetic formation and architecture, their biological functions differ: TCRs bind to peptide antigens presented by the major histocompatibility complex (MHC), and their binding is known to be polyspecific (Rudolph *et al.*, 2006). Antibodies on the other hand can bind to virtually any antigen with high specificity and affinity (Sela-Culang *et al.*, 2013). This distinction in binding preferences may be related to differences in their binding site structures (Wong *et al.*, 2019). For both TCRs and antibodies, the majority

of the binding site is made up of six loops known as the complementarity-determining regions (CDRs). It has been shown that when multiple solved structures of an identical sequence TCR CDR are available, 20% adopt distinct conformations, which is far higher than the ~8% observed in antibodies despite the far larger number of antibody structures (Wong *et al.*, 2019). This structural variability may allow alternative binding configurations for a given TCR, leading to its polyspecificity (Garcia *et al.*, 1998; Morris and Allen, 2012; Wong *et al.*, 2019).

Several packages exist to model the structures of TCRs including LYRA, TCRModel and Repertoire Builder (Klausen *et al.*, 2015; Gowthaman and Pierce, 2018; Schmitt *et al.*, 2019). However, none of these tools produce predictions that reflect the multiple conformations of TCR CDRs. Here, we present TCRBuilder, a multi-state TCR modelling server. TCRBuilder is based on the ABodyBuilder pipeline for antibody modelling (Leem *et al.*, 2016) and predicts TCR models with multiple CDR conformations where applicable.

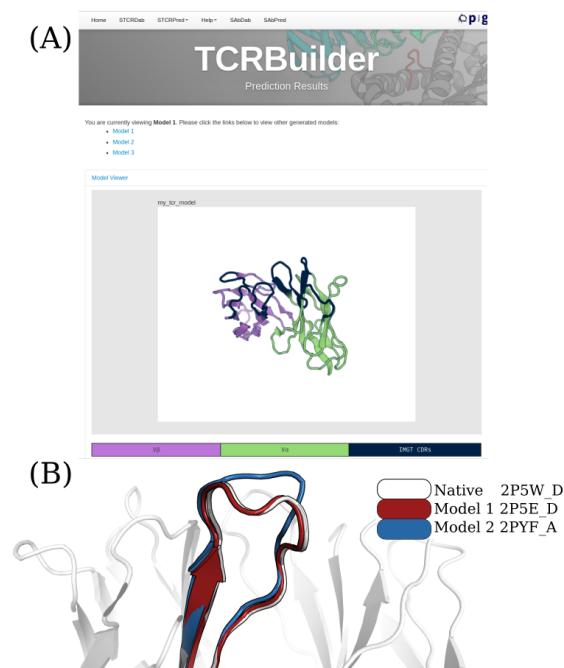


Fig. 1. (A) An example result page of the TCRBuilder web server. (B) An example case where a CDR α 3 sequence (AVRPLLDGTYIPT) with multiple distinct conformations was modelled. One of the conformations (2P5E chain D in red) was nearly identical to the native 2P5W chain D structure, but an alternative conformation of this loop (2PYF chain A in blue) is also predicted as a possible shape.

2 Webserver

TCRBuilder is available as a web server (Fig. 1A). It takes as input a paired $\alpha\beta$ TCR sequence, and returns model structures with simple biophysical property annotations similar to those in ABodyBuilder (Leem *et al.*, 2016). The detailed pipeline is described in the Supplementary Materials.

2.1 Performance analysis

To evaluate TCRBuilder, we carried out a leave-one-out cross-validation on 169 non-redundant TCR sequences in STCRDab (Leem *et al.*, 2018), barring the algorithm from using full sequence-identical template structures. The performance was assessed by superimposing the predicted model onto the native structure, and calculating the root-mean-square-distance (RMSD) over the backbone atoms of the given region. RMSDs over the individual CDRs were obtained after aligning two anchor residues before and two after the CDR. If there were multiple crystal structures or models for a given sequence, all measurements were recorded. The average RMSD for the complete model was 1.49 Å. In the models, CDRs 1 and 2 on both the α and β chains are well predicted, but CDR3 less so. Of the 169 models generated, 106 required template-free modelling (Marks *et al.*, 2017) for at least one of their CDRs. This is a far larger percentage than the <4% of antibody cases that required template-free modelling in a large test (Leem *et al.*, 2016). These results highlight the lack of structural data available for TCRs.

To test whether TCRBuilder is comparable to current state-of-the-art, freely-available TCR webserver, we constructed a blind test set of the 10 (9 non-redundant) $\alpha\beta$ TCR structures released between 24 July 2019 and 22 October 2019. We compared the modelling performance of LYRA, TCRModel and Repertoire Builder with TCRBuilder (Klausen *et al.*, 2015; Gowthaman and Pierce, 2018; Schmitt *et al.*, 2019), preventing them from using any structures released since 24 July 2019. TCRBuilder

has comparable performance with existing TCR modelling tools, on this set of unseen structures (Supplementary Materials).

2.2 Predicting multiple conformations

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3 Conclusion

TCRBuilder is the first TCR modelling server that captures the multiple stable conformations in TCR CDRs. The multi-state models it generates better represent the possible binding configurations a TCR can adopt, paving the way to understand polyspecificity in TCRs.

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